Supplementary Materials for: A Two-Stage Clustered MILP Framework for Efficient and Resilient Waste Collection Routing

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1. Introduction to the Supplementary Materials

This document extends the paper "A Two-Stage Clustered MILP Framework for Efficient and Resilient Waste Collection Routing" (ICERIE 2027, Paper ID 87). The paper develops a cluster-first, route-second framework for the Rajshahi City Corporation (RCC). K-means groups the secondary transfer stations (STS) into zones, and a mixed-integer linear programming (MILP) model then routes a heterogeneous fleet of shuttles and sweepers inside each zone. The page limit of the paper left no room for the full data table, the cluster assignments, the complete model with units and explanations, the scenario results at full precision, or the extended economic analysis. They are collected here so that readers can inspect and reproduce the work.

Values marked as derived were obtained by simple addition or division of tabulated numbers. All other values are reported as used in the study. Equations are numbered as in the paper, with Eq. (8) and Eq. (9) written out in two parts each.

Abbreviations. BDT: Bangladeshi taka; CO_2: carbon dioxide; CVRPTW: capacitated vehicle routing problem with time windows; MILP: mixed-integer linear programming; MTZ: Miller–Tucker–Zemlin; RCC: Rajshahi City Corporation; STS: secondary transfer station.

2. Supplementary Dataset and STS Information

2.1 Data sources

Data were collected from 20 STS located across Rajshahi. They cover residential, commercial, institutional and industrial areas. Each STS record holds its latitude, longitude and estimated daily waste generation in tons per day. Primary data came from field observations. Secondary data came from existing records and the literature. STS information was taken from the assessment of STS in RCC by Ahmad et al. Coordinates and average traffic speeds were obtained from Google Maps Platform (accessed 21 March 2026).

2.2 Vehicle fleet and operational parameters

The fleet has two sweeper classes and one shuttle class. Table S1 lists their specifications and Table S2 lists the operational and environmental parameters. Larger vehicles have higher fuel cost per kilometer, higher driver wages and higher emission factors, so the model has to balance cost, capacity and emissions when it assigns vehicles.

Table S1. Vehicle fleet specifications. Capacity Q_k in tonnes, fuel cost C_k^{fuel} in BDT/km, daily driver wage C_k^{wage} in BDT/day, emission factor E_k in kg CO_2/km.

Class	Role (set)	Q_k (t)	C_k^{fuel}	C_k^{wage}	E_k
Type 1 (1.5 T)	Sweeper (K_R)	1.5	18	1200	0.15
Type 2 (3.0 T)	Sweeper (K_R)	3.0	25	1500	0.22
Type 3 (5.0 T)	Shuttle (K_S)	5.0	35	1800	0.35

Table S2. Operational and environmental parameters.

Parameter	Symbol	Value	Source / description
Congestion factor	β	1.3–1.5	Based on Rajshahi traffic audits
Maximum working time	$T_{\max}$	8 hours	Standard labour shift
Average speed	v	20 km/h	Typical city dumper speed
Carbon penalty	λ	3.85 BDT/kg CO_2	Cramton and Stoft (2009)

The congestion factor β adjusts both travel time and fuel consumption: the effective speed is v/β. The sensitivity analysis in Section 6 uses $\beta = 1.4$ and $\lambda = 3.85$ BDT/kg CO_2 as the baseline.

2.3 STS data

Table S3 gives the name, coordinates, K-means cluster and daily waste generation of each STS. The index n is the label printed on the corresponding point in Figure S1. Daily generation ranges from 8 t/day at Haragram STS to 31 t/day at RMC STS. Other high-generation stations are Mission Hospital STS (29 t/day) and RU STS (28 t/day). Airport STS generates 11 t/day.

Table S3. STS names, coordinates (decimal degrees), cluster assignment and daily waste generation (tons/day).

n	STS name	Latitude	Longitude	Cluster	Waste (t/day)
1	Kashiadanga STS	24.3736	88.5778	2	9
2	Adorsho School STS	24.3705	88.5852	2	13
3	Court Walton STS	24.3718	88.5895	2	21
4	New Degree College STS	24.3689	88.5938	2	27
5	Mission Hospital STS	24.3725	88.5962	0	29
6	STS Bondhogate	24.3732	88.5980	0	17
7	Terokhadia STS	24.3768	88.6015	0	13
8	Match Factory STS	24.3785	88.6068	0	24
9	STS near Sericulture	24.3738	88.6035	0	20
10	RMC STS	24.3729	88.6052	0	31
11	TTC STS	24.3702	88.6020	0	14
12	STS Borokuthi	24.3680	88.6005	0	18
13	Tikapara STS	24.3708	88.6115	0	25
14	Kazla STS	24.3695	88.6355	3	27
15	RU STS	24.3678	88.6288	3	28
16	Mansur Ali Park STS	24.3755	88.6102	0	20
17	Nowdapara STS	24.3925	88.6065	0	15
18	Haragram STS	24.3765	88.5616	2	8
19	Sopura STS	24.3660	88.5880	2	13

n	STS name	Latitude	Longitude	Cluster	Waste (t/day)
20	Airport STS	24.4372	88.6160	1	11

3. Detailed Clustering Methodology and Results

3.1 Method

K-means clustering (MacQueen, 1967; Sinaga and Yang, 2020) partitions the STS by their geographic coordinates, latitude and longitude. It minimizes the within-cluster sum of squared distances, so STS that lie close together end up in the same cluster. Each cluster is an isolated routing zone served by its own sub-fleet. This removes overlapping routes between zones and shrinks the routing problem that the MILP has to solve. The MILP of Section 4 is solved separately for each cluster. The study used $K = 4$ clusters, labeled Cluster 0 to Cluster 3.

3.2 Cluster assignments and characteristics

Cluster 0 is the largest, with 11 STS in the dense central and northern part of the city. Cluster 2 covers six STS along the western side. Cluster 3 holds the two eastern stations, Kazla and RU. Cluster 1 contains only Airport STS, which lies far to the north of all other stations. Table S4 lists the members. The number of STS and the daily waste per cluster are derived by addition from Table S3.

Table S4. Cluster membership. Count and waste sum are derived from Table S3.

Cluster	STS (n)	Members	Count	Waste sum (t/day)
0	5–13, 16, 17	Mission Hospital, Bondhogate, Terokhadia, Match Factory, Sericulture, RMC, TTC, Borokuthi, Tikapara, Mansur Ali Park, Nowdapara	11	226
1	20	Airport	1	11
2	1–4, 18, 19	Kashiadanga, Adorsho School, Court Walton, New Degree College, Haragram, Sopura	6	91
3	14, 15	Kazla, RU	2	55
All	1–20		20	383

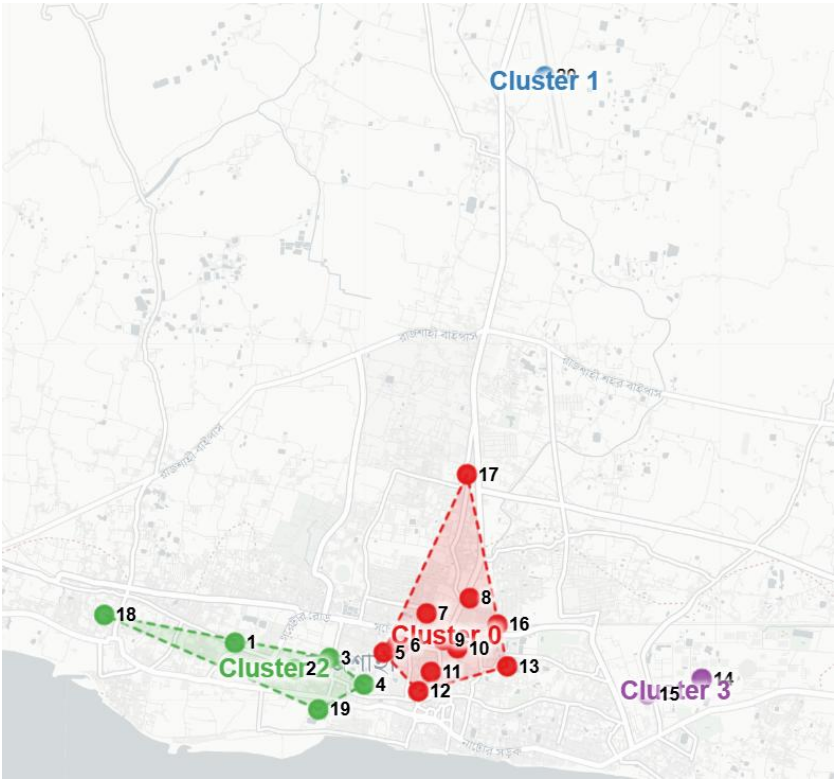


Figure S1. Clustering of the 20 STS into four clusters by K-means on a street map of Rajshahi. Point labels are the indices n of Table S3. Cluster 0 is red, Cluster 1 blue, Cluster 2 green and Cluster 3 purple.

4. Complete Mathematical Formulation of the Two-Stage MILP Model

The model is written for a single cluster. It is a capacitated vehicle routing problem with time windows (CVRPTW) with a heterogeneous fleet of direct-trip shuttles and multi-stop sweepers.

4.1 Sets and indices

- $i, j \in \mathcal{N}$: all network nodes in the cluster. Node 0 is the central depot (landfill) and $\mathcal{C} = \mathcal{N} \backslash \{0\}$ is the set of STS nodes.
- $k \in K_S$: shuttle vehicles, which extract bulk waste by direct depot-to-STS round trips.
- $k \in K_R$: sweeper vehicles, which collect residual waste along multi-stop tours.
- $m \in M = \{1,2, \ldots, |M|\}$: trip slots. $|M|$ is the maximum number of sequential tours a sweeper can complete in one shift. It is fixed beforehand as an upper bound from the waste volume of the cluster and the capacity of the assigned fleet, so the solver does not generate trips during the solve.
- $\mathcal{A}$: admissible arcs, $\mathcal{A} = \left\{(i,j) \;\middle|\; \frac{2D_{0i}}{v/\beta} + t_{LU}^{R} \leq T_{\max},\ i \neq j\right\}$.

4.2 Parameters

Table S5. Model parameters with units.

Symbol	Definition	Unit
Q_k	Payload capacity of vehicle k	t
C_k^{fuel}	Fuel cost per km of vehicle k	BDT/km

Symbol	Definition	Unit
C_k^{wage}	Fixed daily labour cost of vehicle k	BDT/day
E_k	CO_2 emission factor of vehicle k	kg CO_2/km
λ	Carbon penalty rate	BDT/kg CO_2
$\bar{S}_i$	Nominal waste accumulation at STS i	t
$\hat{S}_i$	Maximum surge demand above the nominal level at STS i	t
Γ	Robustness budget, $\Gamma \in [0,1]$	none
D_{0i}	Road-network distance from the depot to STS i	km
D_{ij}	Road-network distance from node i to node j	km
v	Free-flow travel speed	km/h
β	Congestion multiplier, $\beta \geq 1$; effective speed is v/β	none
t_{LU}	Loading and unloading service time per stop	h
$[E_i, L_i]$	Hard time window at STS i: earliest and latest permitted arrival	h
B_{dur}	Mandatory driver rest per deployed vehicle	h
$T_{\max}$	Maximum shift duration	h
$\mathcal{M}$	Dynamic big-M constant for the time-continuity linearization	h

4.3 Decision variables

Table S6. Decision variables.

Variable	Meaning
$y_{ik} \in \mathbb{Z}_{\geq 0}$	Number of direct depot-to-STS round trips made by shuttle k to STS i per shift
$x_{ijkm} \in \{0,1\}$	1 if sweeper k uses arc (i,j) on trip m, otherwise 0
$q_{ikm} \geq 0$	Residual waste (t) collected by sweeper k at node i on trip m
$T_{ikm} \geq 0$	Arrival time (h from shift start) of sweeper k at node i on trip m
$u_{ikm} \geq 1$	Auxiliary MTZ visit-order variable for sweeper k at node i on trip m
$N_k \in \mathbb{Z}_{\geq 0}$	Number of physical vehicles of type k deployed

4.4 Objective function

The objective minimizes the total daily operating cost, which is the sum of fuel, wage and carbon costs.

$$\min Z = Z^{Fuel} + Z^{Wages} + Z^{Carbon} \qquad (1)$$

The wage cost is a fixed daily labour cost for every deployed vehicle and does not depend on distance.

$$Z^{Wages} = \sum_{k \in K_S \cup K_R} N_k \; C_k^{wage} \qquad (2)$$

The fuel cost is proportional to distance and is scaled by the congestion factor. A shuttle covers $2D_{0i}$ on each round trip to STS i. A sweeper covers the arcs it uses.

$$Z^{Fuel} = \sum_{k \in K_S} \beta \; C_k^{fuel} \left[\sum_{i \in \mathcal{C}} 2 \, D_{0i} \, y_{ik} \right] + \sum_{k \in K_R} \sum_{m \in M} \beta \; C_k^{fuel} \left[\sum_{(i,j) \in \mathcal{A}} D_{ij} \; x_{ijkm} \right] \qquad (3)$$

The carbon cost applies the penalty rate to the emissions over the same distances.

$$Z^{Carbon} = \sum_{k \in K_S} \lambda E_k \left[\sum_{i \in \mathcal{C}} 2 D_{0i} y_{ik} \right] + \sum_{k \in K_R} \sum_{m \in M} \lambda E_k \left[\sum_{(i,j) \in \mathcal{A}} D_{ij} x_{ijkm} \right] \quad (4)$$

4.5 Constraints

Every STS receives at least one shuttle round trip per shift, and the combined shuttle and sweeper service must cover the worst-case demand under the robustness budget Γ.

$$\sum_{k \in K_S} y_{ik} \geq 1, \quad \forall i \in \mathcal{C} \quad (5)$$

$$\sum_{k \in K_S} Q_k y_{ik} + \sum_{k \in K_R} \sum_{m \in M} q_{ikm} \geq \bar{S}_i + \Gamma \hat{S}_i, \quad \forall i \in \mathcal{C} \quad (6)$$

Flow balance keeps each sweeper trip continuous. Each trip leaves the depot at most once and visits each STS at most once, which prevents route fragmentation.

$$\sum_{j \in \mathcal{N}} x_{ijkm} = \sum_{j \in \mathcal{N}} x_{jikm}, \quad \forall i \in \mathcal{N},\ k \in K_R,\ m \in M \quad (7)$$

$$\sum_{j \in \mathcal{C}} x_{0jkm} \leq 1, \quad \forall k \in K_R,\ m \in M \quad (8a)$$

$$\sum_{j \in \mathcal{N}} x_{jikm} \leq 1, \quad \forall i \in \mathcal{C},\ k \in K_R,\ m \in M \quad (8b)$$

Collected waste is linked to visitation and bounded by vehicle capacity. Unused trips collect nothing.

$$q_{ikm} \leq Q_k \sum_{j \in \mathcal{N}} x_{ijkm}, \quad \forall i \in \mathcal{C},\ k \in K_R,\ m \in M \quad (9a)$$

$$\sum_{i \in \mathcal{C}} q_{ikm} \leq Q_k \sum_{j \in \mathcal{C}} x_{0jkm}, \quad \forall k \in K_R,\ m \in M \quad (9b)$$

The Miller–Tucker–Zemlin constraint (Desrochers and Laporte, 1991) removes subtours by enforcing a consistent visiting order.

$$u_{ikm} - u_{jkm} + |\mathcal{C}| x_{ijkm} \leq |\mathcal{C}| - 1, \quad \forall i,j \in \mathcal{C},\ i \neq j,\ k \in K_R,\ m \in M \quad (10)$$

Hard time windows apply only at visited nodes. Arrival times propagate along active arcs through a big-M linearization, and the big-M constant is set tightly to improve numerical stability and the strength of the linear relaxation.

$$E_i \sum_{j \in \mathcal{N}} x_{ijkm} \leq T_{ikm} \leq L_i \sum_{j \in \mathcal{N}} x_{ijkm}, \quad \forall i \in \mathcal{C},\ k \in K_R,\ m \in M \quad (11)$$

$$T_{jkm} \geq T_{ikm} + \frac{D_{ij}}{v/\beta} + t_{LU} - \mathcal{M}\left(1 - x_{ijkm}\right), \quad \forall (i,j) \in \mathcal{A},\ j \neq 0,\ k \in K_R,\ m \in M \quad (12)$$

$$\mathcal{M} = T_{\max} + \max_{(i,j)\in\mathcal{A}} \left[\frac{D_{ij}}{v/\beta} + t_{LU}\right] \qquad (13)$$

The total operating time of a sweeper (travel, service and mandatory rest) cannot exceed the shift length. An analogous limit applies to the cumulative service time of a shuttle.

$$\sum_{m\in M} \sum_{(i,j)\in\mathcal{A}} \left[\frac{D_{ij}}{v/\beta} + t_{LU}\right] x_{ijkm} + B_{dur} \le T_{\max}, \quad \forall k \in K_R \qquad (14)$$

$$\sum_{i\in\mathcal{C}} y_{ik} \left[\frac{2D_{0i}}{v/\beta} + t_{LU}\right] + B_{dur} \le T_{\max}, \quad \forall k \in K_S \qquad (15)$$

Finally, the number of deployed vehicles is bounded from below by the activated trips for sweepers, and by the cumulative round trips for shuttles, which are limited to five feasible trips per shift.

$$N_k \ge \sum_{j\in\mathcal{C}} x_{0jkm}, \quad \forall k \in K_R, m \in M \qquad (16)$$

$$N_k \ge \frac{1}{5} \sum_{i\in\mathcal{C}} y_{ik}, \quad \forall k \in K_S \qquad (17)$$

4.6 Assumptions

1. All STS are fully operational throughout the planning horizon, and the depot has no capacity limit.
2. Vehicles of the same class have identical fuel efficiency and emission rates.
3. Travel speed is constant on all arcs within a scenario.
4. The depot supplies as many vehicles and drivers as the daily demand requires.

4.7 Solution approach

The study follows a cluster-first, route-second strategy. In the first stage, K-means partitions the STS by their coordinates. In the second stage, the MILP above is solved for each cluster with Gurobi. Shuttles move bulk waste between the depot and an STS, and sweepers collect the residual waste along multi-stop routes. The routing plan was then tested on eleven demand scenarios (Z_0 to Z_{10}) and in a sensitivity analysis. Figures S3 and S4 show the routing structure inside a cluster and the overall workflow.

5. Detailed Optimization Results

5.1 Traditional and optimized daily cost by scenario

Eleven scenarios were evaluated. Z_0 is the baseline (observed) case and Z_1 to Z_{10} are simulated stochastic demand scenarios. The traditional cost is an estimate for fixed-route, direct depot-to-STS dispatch with no route optimization and no demand uncertainty. The optimized cost comes from the cluster-wise MILP with robust demand buffering ($\Gamma = 0.2$), time-window constraints and multi-shift scheduling. Table S7 gives the results at full precision.

Table S7. Daily collection cost of traditional dispatch and the optimized MILP solution. Waste in tons per day; costs in BDT/day.

Scenario Z_n	Waste (t)	Traditional (BDT)	Optimized (BDT)	Saved (BDT)	Reduction (%)
0	459.60	119,723.35	105,223.67	14,499.68	12.11

Scenario Z_n	Waste (t)	Traditional (BDT)	Optimized (BDT)	Saved (BDT)	Reduction (%)
1	441.69	117,160.22	96,636.83	20,523.39	17.52
2	470.60	119,343.62	104,943.62	14,400.00	12.07
3	440.88	112,451.35	98,531.63	13,919.73	12.38
4	441.41	115,601.46	98,114.24	17,487.22	15.13
5	452.90	121,108.08	101,720.54	19,387.54	16.01
6	470.52	121,422.26	104,902.52	16,519.74	13.61
7	463.63	117,123.26	104,523.26	12,600.00	10.76
8	452.16	113,738.94	96,815.05	16,923.90	14.88
9	434.68	115,911.26	100,471.30	15,439.95	13.32
10	439.72	120,810.39	100,616.72	20,193.67	16.72

The cost reduction ranges from 10.76% (Z_7) to 17.52% (Z_1), and the daily saving ranges from 12,600 BDT to 20,523.39 BDT. Total savings divided by total traditional cost gives an average reduction of 14.05% (derived). Reductions stay in a narrow band while the total waste moves between 434.68 and 470.60 t/day.

5.2 Route allocation and fleet information

The optimized solution replaces the conventional radial dispatch with a multi-trip routing network in each cluster. Its effect is a smaller fleet and less redundant travel. Route sequences, the number of vehicles deployed per class and cluster (N_k) and route distances are not tabulated in this document.

6. Supplementary Economic Analysis

6.1 Economic comparison

Figure S2 plots the daily cost of the traditional and optimized systems for Z_0 to Z_{10}. The exact values are in Table S7.

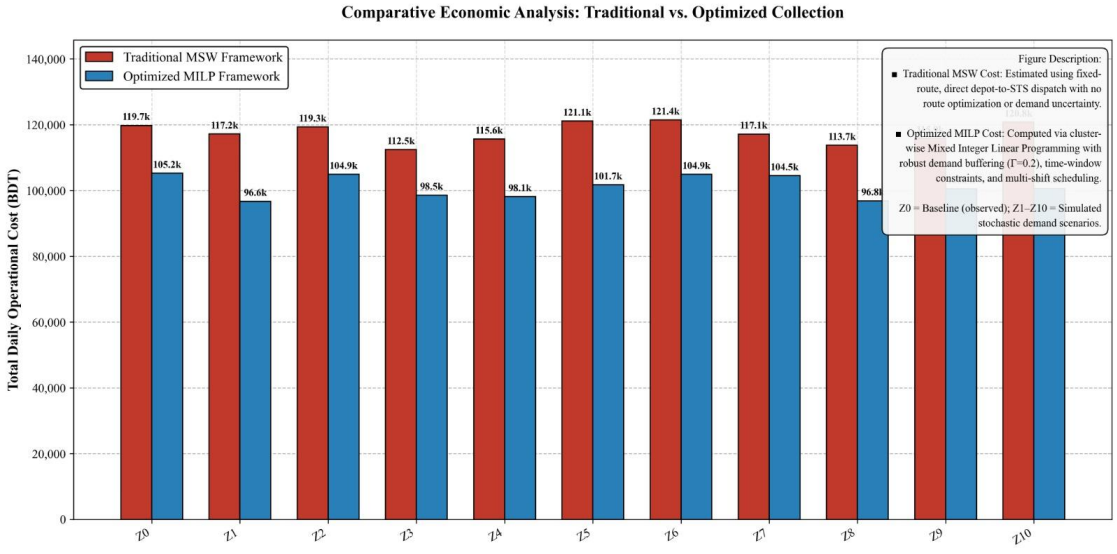


Figure S2. Comparative economic analysis: traditional versus optimized collection. Total daily operational cost in BDT for scenarios Z_0 to Z_{10}.

6.2 Sensitivity analysis

The carbon tax λ, the fuel price and the congestion factor β were varied one at a time around the baseline ($\lambda = 3.85$ BDT/kg CO_2, fuel price 100%, $\beta = 1.4$). Table S8 gives the resulting total daily cost Z. The change columns are derived: for λ relative to $\lambda = 0$, for fuel price relative to the 100% row, and for β relative to $\beta = 1.4$.

Table S8. Total daily cost Z (BDT) under variation of carbon tax λ (BDT/kg CO_2), fuel price level and congestion factor β.

λ	Z	**Change (%)**
0	95,406.160	0
2	96,249.323	+0.88
3.85 (baseline)	97,036.715	+1.71
6	97,954.351	+2.67
8	98,811.913	+3.57

Fuel price	Z	**Change (%)**
80%	86,581.190	−12.37
90%	92,693.720	−6.19
100% (baseline)	98,806.250	0
110%	104,918.781	+6.19
120%	111,031.312	+12.37

β	Z	**Change (%)**
1.0	79,541.878	−19.50
1.2	88,274.064	−10.66
1.4 (baseline)	98,806.251	0
1.6	107,471.296	+8.77
1.8	117,914.829	+19.34

Over the tested ranges, raising the carbon tax from 0 to 8 BDT/kg CO_2 increases the cost by 3.57%. Raising the fuel price from 80% to 120% of the baseline increases it by 28.24%, and raising β from 1.0 to 1.8 increases it by 48.24%.

6.3 Combined effect of congestion and fuel price

In the two-way analysis, the cost ranges from 70.8k BDT ($\beta = 1.0$, lowest fuel price level) to 134.6k BDT ($\beta = 1.6$, highest fuel price level). These are 28.3% below and 36.2% above the 98.8k BDT baseline. The highest fuel price level alone ($\beta = 1.4$) raises the cost to 123.0k BDT (+24.5%). $\beta = 1.6$ alone raises it to 107.5k BDT (+8.8%). Together they raise it to 134.6k BDT (+36.2%).

6.4 Interpretation

Congestion and fuel price are the main operational risk factors. Congestion lengthens travel time, raises fuel consumption and lowers vehicle utilization. Fuel price acts directly on transport cost. The carbon tax adds about 3.6% over the tested range, so it mainly encourages environmentally sustainable transport choices, while traffic management and fuel optimization offer larger economic gains. Because congestion

and fuel price compound at the upper end of the tested range, dynamic re-optimization suits the problem better than a static daily budget.

7. Additional Figures and Visualizations

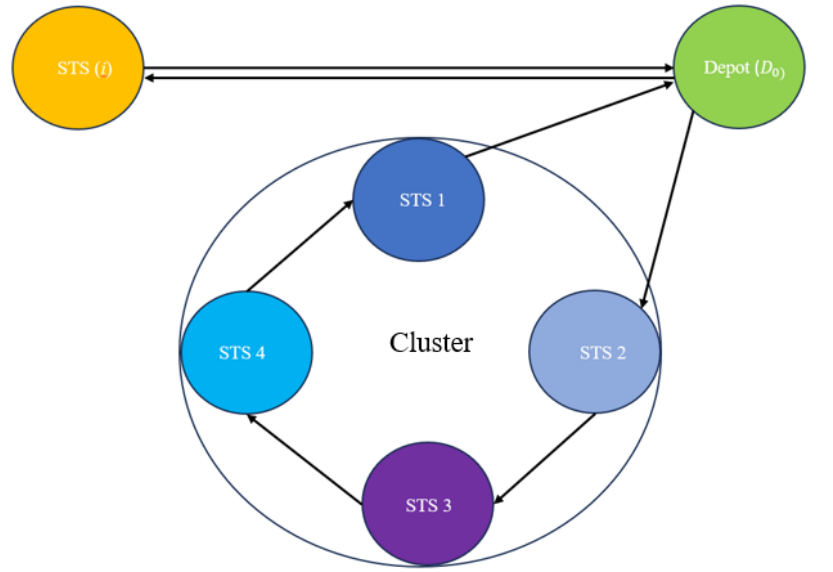


Figure S3. Vehicle routing in a single cluster. A sweeper leaves the depot, visits several STS in sequence and returns to the depot.

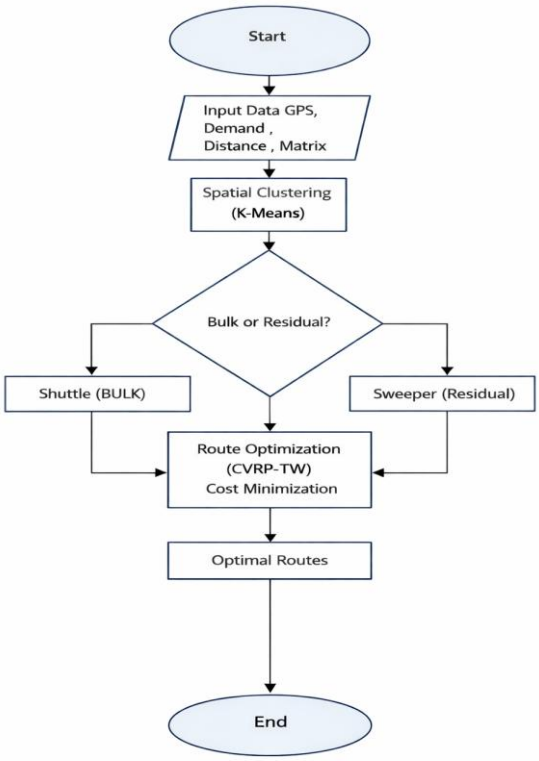


Figure S4. Flowchart of the overall process: input data, K-means clustering, the bulk or residual decision, route optimization by the CVRPTW model, and optimal routes.

8. Reproducibility Notes and Data Availability

8.1 Inputs not tabulated in this document

The model inputs listed below are not tabulated here and are needed to rerun the experiments: the depot coordinates and the distance matrices D_{0i} and D_{ij}; the values of t_{LU}, B_{dur}, $|M|$, the time windows $[E_i, L_i]$ and the surge demand $\hat{S}_i$; the procedure used to generate the scenarios Z_1 to Z_{10}; the K-means settings (initialization, random seed, number of restarts); and the Gurobi version, time limit and optimality gap.

8.2 Data and code availability

The numerical data of this document are in Tables S1 to S8. The STS dataset (Table S3) and the cost tables (S7 and S8) can be released as CSV files alongside this document. A repository DOI or URL will be added once the repository is created.

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